



Airway, Drug Assisted (OPTIONAL)

Indications for Drug Assisted Airway

Failure to protect the airway
and/or
Unable to oxygenate
and/or
Unable to ventilate
and/or
Impending airway compromise

Capnography Monitoring

- End-tidal (EtCO₂) monitoring is mandatory following placement of an endotracheal tube.
- EtCO₂ monitoring is mandatory following placement of a BIAID once available on scene.

Protocols AR 1, 2, 3, 5, and 6 should be utilized together (even if agency is not using Drug Assisted Airway Protocol) as they contain useful information for airway management.

	Preoxygenate 100% O ₂
	IV or IO Access Protocol UP 6 2 points of access
P	Assemble Airway Equipment Suction equipment Alternative Airway Device

Hypoxic Or
Hypotensive Or
Dangerously Combative?

YES

NO

P	Etomidate 0.3 mg/kg IV / IO Or Ketamine 1 - 2 mg/kg IV / IO May repeat x 1
	Succinylcholine 2 mg/kg IV / IO Or Rocuronium 1 mg/kg IV / IO (if Succinylcholine contraindicated) May repeat x 1
	Intubate trachea
	Placement Verified Continuous Waveform Capnography

Consider Restraints Physical Procedure

P Consider Gastric Tube Insertion Procedure

Awakening or Moving
after intubation

NO

Exit to
Post-intubation/ BIAID Management
Protocol AR 8

YES

P	Ketamine 1 - 2 mg/kg IV / IO No IV or IO Access: Ketamine 4 mg/kg IM Maximum 400 mg Pediatric: See Pearls 2 mg/kg IM Maximum 400 mg
	Correct Hypoxia and/ or Hypotension
	Age Appropriate Airway Protocol(s) AR 1, 2, 5, 6, as indicated
	Hypotension/ Shock Protocol AM 5/ PM 3 as indicated

Hypoxia corrected
Hypotension corrected
Dangerously Combative condition corrected

YES

Patient still requires intubation

NO

Exit to
Appropriate
Protocol(s)

Procedure will remove
patient's protective
airway reflexes and
ability to breath.

You must be sure of
your ability to intubate
before beginning this
procedure.

Must have two (2)
Paramedics on scene

Red Text

Key performance
indicators used to
evaluate protocol
compliance.

A NCOEMS Airway
Evaluation Form must
be completed for every
patient who receives
Drug Assisted Airway.



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Pearls

- Agencies must maintain a separate Performance Improvement Program specific to Drug Assisted Airway.
- This procedure requires at least 2 Paramedics. See Pearls section of protocols AR 1 and 2.
- For the purposes of this protocol, a secure airway is when the patient is receiving appropriate oxygenation and ventilation.
- If an effective airway is being maintained by BVM with continuous pulse oximetry values of $\geq 90\%$, it is acceptable to continue with basic airway measures.
- Ventilation rate:
30 for Neonates, 25 for Toddlers, 20 for School Age, and for Adolescents the normal Adult rate of 10 - 12 per minute. Maintain EtCO₂ between 35 - 45 and avoid hyperventilation.
- Hypoxia and/ or Hypotension:
Increased risk of cardiac arrest when a sedative with paralytic medications are administered while hypoxic and/ or hypotensive. Resuscitation and correction of hypoxia and/ or hypotension are paramount prior to use of these combined agents. Ketamine administration allows time for appropriate resuscitation of hypoxia and/or hypotension while managing the airway.
- Ketamine for airway intervention and/ or sedation purposes:
Ketamine may be used in pediatric patients (fit within a Pediatric Medication/ Skill Resuscitation System product, ≤ 15 years of age, or ≤ 49 kg) with DIRECT ONLINE MEDICAL ORDER by the system MEDICAL DIRECTOR or ASSISTANT MEDICAL DIRECTOR only.
Agencies using Ketamine in the pediatric population must also be using in their adult population.
- KETAMINE:
Ketamine may be used with or without a paralytic agent in conjunction with either an OPA, NPA, BIAD or endotracheal tube. (BIAD is preferred over endotracheal tube until hypoxia and/ or hypotension are corrected). Ketamine may be used during the resuscitation of hypoxia or hypotension in conjunction with airway management. Once hypoxia and hypotension are corrected, use of a sedative and paralytic can proceed if indicated. Ketamine may be used in the dangerously combative patient requiring airway management IM. IV/ IO should be established as soon as possible. Ketamine may be used for sedation once a BIAD or endotracheal tube are established and confirmed. Agencies using Ketamine must follow Standards Policy: Medical Policy Section Ketamine Program Requirements. Medical Policy 2.
- Intubation attempt defined as laryngoscope blade passing the teeth or endotracheal tube passed into the nostril.
- If First intubation attempt fails, make an adjustment and try again: (Consider change of provider in addition to equipment)
- NC EMS Airway Evaluation Form:
Fully complete and have receiving healthcare provider sign confirming BIAD or endotracheal tube placement. Complete online in region specific ReadyOp and upload completed form. Complete when Ketamine, Etomidate, Succinylcholine and/ or Rocuronium or used to facilitate use of a BIAD and/ or endotracheal intubation.
- Paramedics/ AEMT should consider using a BIAD if endotracheal intubation is unsuccessful.
- Drug Assisted Airway is not recommended in an urban setting (short transport) when able to maintain oxygen saturation $\geq 90\%$.
- **DOPE:** Displaced tracheostomy tube/ ETT, Obstructed tracheostomy tube/ ETT, Pneumothorax and Equipment failure.